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The Palaeomagnetism of the Antrim Plateau Volcanics of Northern Australia

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Summary

The magnetization of samples from 29 sites in Lower Cambrian Volcanics and interbedded sediments, collected over an area of 250 000 km^2 of northern Australia, has been investigated. Fourteen of the sites respond to magnetic cleaning, although in one case thermal cleaning is preferred. The other sites either become randomly magnetized or remain magnetized in the direction of the present geomagnetic field, as in the case of the interbedded sediments The mean direction of $D = 51°$, $I = +66°$ ($\alpha_{95} = 13°$) gives a palaeomagnetic pole at 9 N, 160 E with $A_{95} = 17°$ The polar-wander path for Australia is for the first time extended back to Lower Cambrian times with confidence, and a complete curve can now be drawn for the Phanerozoic. It is concluded that Australia was situated in the Northern Hemisphere in Cambrian times. Two rapid polar transitions are identified during the Palaeozoic, a feature which is also present in data from Africa. The polar-wander paths for these two continents can now be made to coincide for the whole of the Palaeozoic, implying that their relative positions remained unchanged during this period.

Introduction

The first palaeomagnetic measurements made on Australian Cambrian rocks were reported by Irving & Green (1958), who investigated the natural remanent magnetization (NRM) of samples of Antrim Plateau Volcanics and of some sediments collected by the Bureau of Mineral Resources. Results from the Volcanics were somewhat inconclusive and Irving (1964) notes that the mean direction quoted is based on only seven samples. A further collection of 15 sites was therefore made by Irving (personal communication) in 1958, but these samples failed to respond satisfactorily to alternating field (a.f.) demagnetization, and no results were obtained from this collection.

Because of the importance of obtaining results from the lower Palaeozoic of Australia, the Antrim Plateau Volcanics, as virtually the only basic igneous rocks of this age, possess particular palaeomagnetic significance. Therefore since both the geological mapping of the area covered by the Volcanics and its accessibility have been greatly improved during the past ten years, it was decided to undertake a third collection from this formation in 1968.

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Geology

Plateau basalts in northern Australia extend over about 400 000 square kilometres in an area between the Kimberley Region of Western Australia and the Queensland border (Dunn & Brown 1969). The Antrim Plateau Volcanics incorporate most of these basalts and crop out extensively in the East Kimberley, Victoria River and Daly River Basin areas (Fig. 1). The stratigraphic relationships of the Volcanics in these areas are given in Table 1, which separates the East Kimberley area north and south into the 'Bonaparte Gulf Basin' and the 'Southern Ord Area', while combining the other two areas as the 'Northern Territory'.

The Volcanics consist predominantly of massive grey, dark green and purple tholeiitic basalt, with fine-grained vesicular basalt near the top and bottom of the flows. Amygdaloidal basalt comprises about one-third of the section and there are

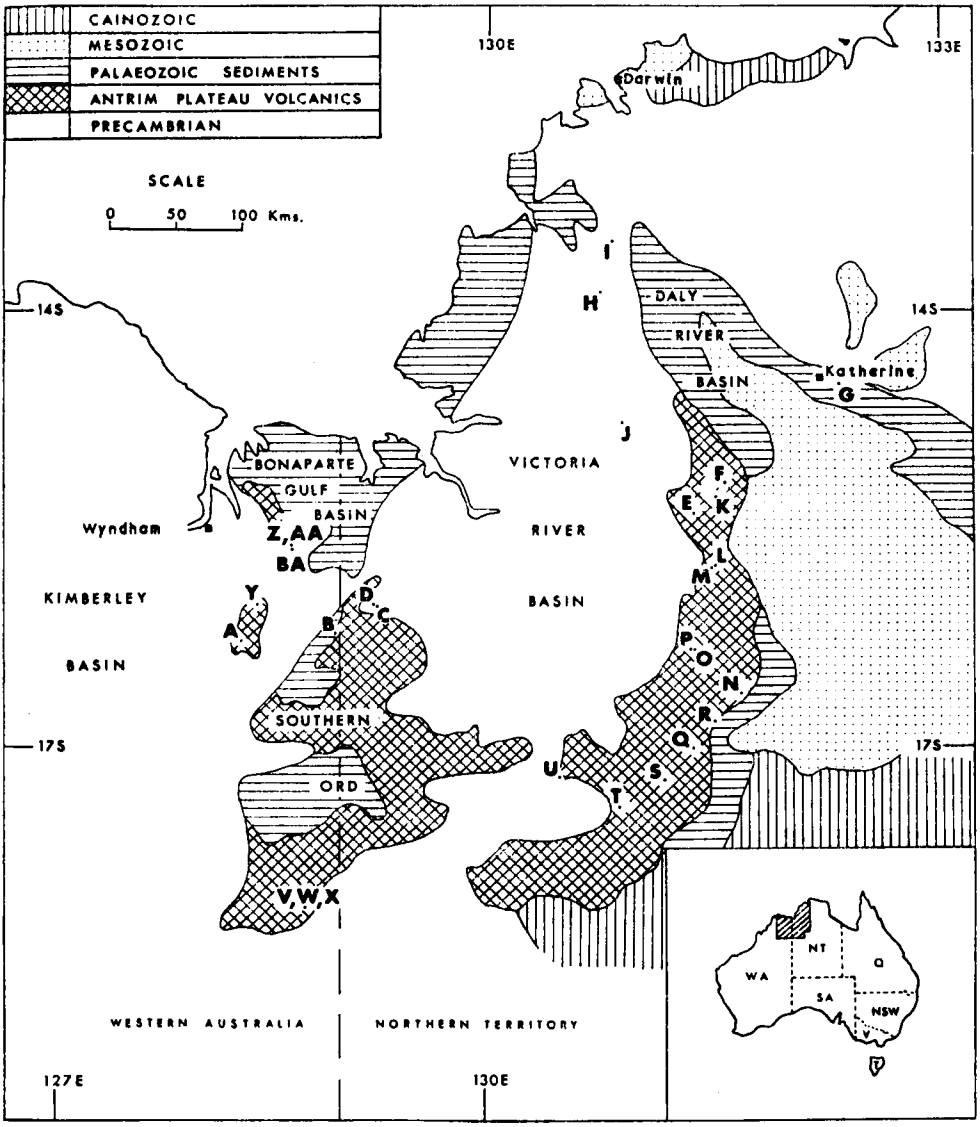


FIG. 1. Geological sketch map of northern Australia, indicating sampling sites in the Antrim Plateau Volcanics.

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Table 1

Pre-Carboniferous stratigraphic relationships in northern Australia

	Bonaparte Gulf Basin	Southern Ord Area	Northern Territory
DEVONIAN	Cockatoo Formation	Elder Sandstone	
LOWER ORDOVICIAN	Pander Greensand		DALY
UPPER CAMBRIAN	Clark Sandstone, etc.		RIVER GROUP*
MIDDLE CAMBRIAN	Hart Spring Sandstone, etc.	NEGRI GROUP	ETC.
LOWER CAMBRIAN	Blatchford Formation		
	ANTRIM PLATEAU VOLCANICS		
PRE-CAMBRIAN	ALBERT EDWARD GROUP inc. Timperley Shale 666±43 My Elvire Formation 653±48 My etc.		VICTORIA RIVER GROUP* ETC.

*Current work by the Bureau of Mineral Resources, Geology and Geophysics, indicates that these terminologies are subject to amendment.

also minor tuff and agglomerate beds, together with interbedded sandstone, chert, limestone, conglomerate and pyroclastics. The total thickness of the formation is estimated to exceed 1000 m to the west and 300 m to the east, individual flows being mostly less than 30 m thick (Dow *et al.* 1964). Interbedded sedimentary rocks indicate that some of the basalts were extruded subaerially and some into shallow water. The evidence therefore suggests that the basalts were extruded in an area near sea level and subject to marine incursions (Randal & Brown 1967).

In general the Volcanics exhibit only very gentle dips and appear tectonically undisturbed. However, their structural history close to the Bonaparte Gulf Basin is one involving some minor activity after extrusion, followed by uplift and erosion prior to the Upper Devonian. Later subsidence was followed by major faulting during the Lower Carboniferous. A manifestation of this activity appears at Tarrara Bar in the Ord River (sites Z and AA, Fig. 1) where the basalt dips eastwards at 30° (Veevers & Roberts 1968; Kaulback & Veevers 1970).

The age of the Antrim Plateau Volcanics must be inferred from the stratigraphic evidence, because no diagnostic fossils have been found in the interbedded sediments. Probably the most definitive age limits are set in the East Kimberley Region, where the Volcanics overlie, with a marked angular unconformity, shales dated at 666±43 My (Bofinger 1967), and have been eroded and faulted before the deposition of the Blatchford formation in the late Lower Cambrian (Kaulback & Veevers 1969). This would suggest an age for the Volcanics of early Lower Cambrian or very latest Precambrian.

Supporting evidence for the younger age is seen elsewhere in the Southern Ord Region, where the Volcanics are overlain with only slight angular unconformity by

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Table 2

Site directions from the Antrim Plateau Volcanics

Site position			NRM					Cleaned							Site pole	
Site	Long. °E	Lat. °S	N(n)	*D*	*I*	*R*	k	T	H	N(n)	*D*	*I*	*R*	k	Long.	Lat.
A	128·2	16·3	2(2)	254	+34	1·99	251	500		2(2)†	087	+59	1·99	78		
									400	2(2)†	082	+61	1·99	5000	176 E	05 S
B	128·9	16·3	2(2)	013	−41	1·96	29		400	2(2)†	020	−40	1·99	92		
C	129·3	16·0	4(4)	037	−68	3·39	5·0		450	2(2)	324	−23	1·94	16		
D	129·3	16·0	2(2)	•		1·35			300	2(5)	•		1·53			
E	131·5	15·4	5(7)	•		3·21		500		4(4)	061	+47	3·69	9·7		
									500	5(10)	035	+25	4·73	15	184 E	45 N
F	131·6	15·2	4(5)	•		3·05		500		4(4)	083	+51	3·76	13		
									400	4(10)	083	+69	3·91	34	170 E	08 S
G	132·4	14·4	9(10)	033	−02	7·67	6·0	500		5(5)	073	+56	4·95	75		
									350	9(18)	061	+54	8·82	44	180 E	14 N
H	130·8	13·9	6(7)	•		3·21		550		4(4)	•		2·70			
									500	6(11)	037	+71	5·89	46	151 E	14 N
I	130·9	13·6	2(3)	•		0·71		500		2(2)	•		1·49			
									400	2(4)	021	+84	1·99	169	136 E	02 S
J	131·0	14·8	2(3)	059	+52	1·99	98	550		2(2)	070	+61	1·99	348		
									200	2(4)	063	+57	1·99	216	177 E	11 N
K1	131·6	15·3	2(3)	227	−73	1·97	35	550		2(2)	•		1·83			
									100	2(6)	273	−47	1·97	34	195 E	10 S
K2	131·6	15·3	3(5)	083	−68	2·68	6·2	300		2(2)	•		1·74			
									300	3(7)	028	−53	2·96	48		
L	131·7	15·8	4(6)	021	−52	3·74	12	550		4(4)	048	+55	3·81	15		
									300	4(8)	049	+67	3·89	28	162 E	12 N
M	131·6	15·7	4(6)	022	−42	3·24	4·0	400		4(4)	•		2·85			
									200	4(8)	006	−36	3·94	50		
N	131·8	16·6	2(3)	•		1·57			400	2(4)	190	−77	1·99	170	136 E	09 N

O	131·5	16·4	2(3)		•		1·86		500		2(2)	107	−62	1·99	92		
										300	**2(5)**	**245**	**−87**	**1·97**	**31**	**137 E**	**14 S**
P	131·5	16·3	4(5)	356		−43	3·84	19	‡	‡	4(14)		‡				
Q	131·3	17·0	4(6)		•		2·11		500		4(4)		•	2·02			
										300	4(8)	354	−43	3·97	98		
R	131·5	16·8	2(2)	021		−62	1·99	106		300	2(3)	014	−50	1·99	97		
S	131·2	17·3	2(3)		•		1·56		‡	‡	2(7)		‡				
T	130·9	17·5	5(7)		•		2·99			**350**	**5(15)**	**005**	**+32**	**4·22**	**5·1**	**140 E**	**55 N**
U	130·5	17·3	4(12)	000		−30	3·80	15	**500**		**4(8)**	**302**	**+68**	**3·92**	**38**	**098 E**	**05 N**
										300	4(12)	334	+30	3·74	12		
V	128·7	18·1	3(3)	052		−56	2·97	60	‡	‡	3(4)		‡				
W	128·7	18·1	3(3)		•		2·26		‡	‡	3(4)		‡				
X	128·7	18·1	3(3)	346		−40	2·84	13	‡	‡	3(4)		‡				
Y	128·4	16·2	4(5)	228		−29	3·49	5·9	450		4(4)	233	−33	3·69	9·6		
										200	**3(7)**	**275**	**−55**	**2·88**	**17**	**185 E**	**13 S**
Z	128·7	15·6	3(4)		•		1·32		400		3(3)		•	1·76			
										200	2(6)		•	1·89			
AA	128·7	15·6	4(5)		•		2·26		400		4(4)		*	1·59			
										250	4(9)	319	−04	3·51	6·1		
BA	128·7	15·7	3(4)	350		−25	2·95	43	450		2(2)	356	−15	1·98	65		
										250	3(6)	346	−21	2·94	31		
	FORMATION MEAN DIRECTION (CLEANED)										**14**	**051**	**+66** (α_{95} = 13°)	**12·786**	**11**		
	FORMATION MEAN OF SITE POLES (CLEANED)										**14**			**11·903**	**6·2**	**160 E** (A_{95}=17°)	**09 N**

N = Number of hand samples or cores.
n = Number of specimens measured.
D = Declination in degrees, east of north.
I = Inclination in degrees, positive downwards.
R = Resultant of N unit vectors.
k = Fisher's precision parameter.
α_{95}, A_{95} = Semi-angles of cones of 95 per cent confidence for mean direction and mean pole respectively.

T = Cleaning temperature normally used (°C).
H = Peak alternating magnetic field normally used (oersted).
† = Two specimens from one core.
• = Directions are random by criteria in text.
‡ = No stable directions were observed during either thermal or alternating field demagnetization.

the Middle Cambrian Negri Group. The apparent weathering profile between the Volcanics and the Negri Group represents a much smaller interval of time than the regional angular unconformity at the base of the Volcanics (Dow & Gemuts 1967). Additional evidence is seen in the Northern Territory where the Volcanics are disconformably overlain by Middle Cambrian sediments (Randal & Brown 1967).

The Antrim Plateau Volcanics are therefore assumed to have been extruded during the early Lower Cambrian.

Sampling

Samples were collected from 26 basalt flows and three interbedded sandstone horizons throughout an area of approximately 250 000 square kilometres, as indicated in Fig. 1. The 29 sites were situated at 20 creeks or rivers, two quarries, two small road cuttings and five other roadside exposures. Two cores, 2·2 cm in diameter and up to 50 cm long, were drilled at some of the sites using a portable diamond coring rig, whilst at the other sites a number of hand samples were collected. Both the samples and cores were oriented using either a Brunton compass or sun-compass clinometer.

Many exposures of the Volcanics are severely cracked and jointed, with weathering effects extending several metres from the surface (an effect which probably accounts for the lack of success in previous investigations). At such places it was impossible, or pointless, to drill long cores, so that the portable diamond coring rig was used exclusively at only 10 sites. At one locality (K), prominent sandstone 'dykes' traverse an exposure of very amygdaloidal basalt. Both the basalt and the sediments were sampled and are referred to as sites K1 and K2 respectively in Table 2. The other sedimentary sites listed are M and Q.

Up to eight cylindrical specimens (2·2 cm × 2·2 cm) were cut from each of the 99 samples and cores, and a total of 296 specimens was investigated.

Laboratory measurements

The direction and intensity of the NRM of at least one specimen per sample were measured using an astatic magnetometer (Collinson *et al.* 1957). The intensities ranged widely from $0{\cdot}8 \times 10^{-6}$ to $8{\cdot}2 \times 10^{-2}$ e.m.u. cm^{-3}, and only the directional results are incorporated in Table 2. Thirteen of the sites can be regarded as being randomly magnetized upon initial measurement. In these cases the resultant vectors were less than the values tabulated by Watson (1956) or Vincenz & Bruckshaw (1960), which are exceeded in only five per cent of cases by a set of randomly directed vectors. Where only two long cores were drilled at a site, the mean directions considered significant are associated with vector resultants which exceed 1·93. At the other 16 sites the directions were still very scattered but related to the direction of the present geomagnetic field.

To remove secondary components, alternating field (a.f.) demagnetization was applied to at least two specimens from each sample, using an apparatus similar to that described by McElhinny (1966). The procedure and criteria for magnetic cleaning were as outlined by McElhinny & Gough (1963). The response of samples to this treatment was quite variable and representative demagnetization curves are shown in Fig. 2(a). The curve for site A changes slope sharply when the direction of the specimen reaches a distinct end-point after removal of a substantial secondary component in low fields. Site J is seen to be extremely stable. The curves for O and U show a fairly rapid decrease in intensity at low fields and this typifies the response of most other sites.

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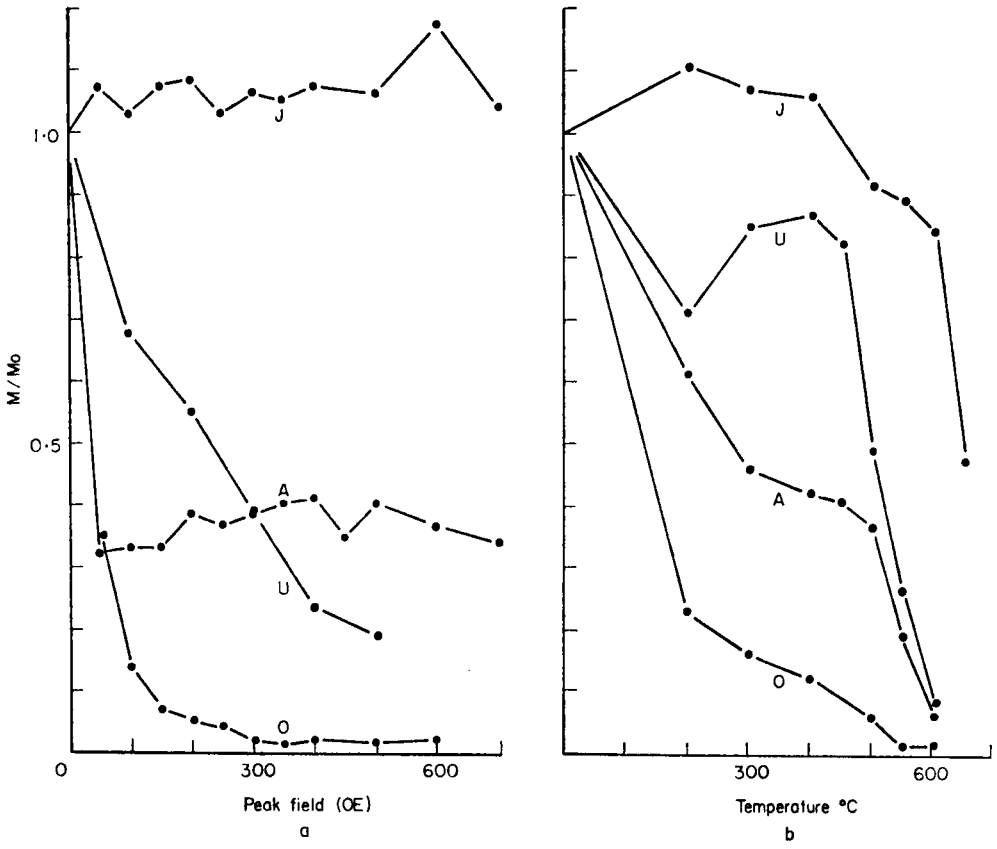


FIG. 2. Demagnetization curves for specimens from four sites, (a) alternating field, (b) thermal.

In general, the site directional results after magnetic cleaning may be classified as follows:

(1) Samples became randomly magnetized;
(2) Sites became randomly magnetized;
(3) Sample directions remained clustered about that of the present geomagnetic field (particularly in the case of the sediments); and
(4) Samples produced significant site mean directions well away from the present geomagnetic field direction.

Results were accepted from 14 sites which fell into category (4). Of these sites, one core from site A and one sample from site Y were in category (1) and were consequently rejected. Consistent directions obtained from specimens at points down the second core from site A have been used in calculating the site mean direction (Table 2).

Because of the variation in response to magnetic cleaning it was decided to thermally demagnetize specimens from each site by successively heating to higher temperatures and cooling in field-free space, using the apparatus described by McElhinny, Luck & Edwards (1970). During each run, the residual field over the volume occupied by the furnace and the specimens was maintained to less than 0·5 gamma. The wide range of thermal demagnetization curves obtained is represented in Fig. 2(b) by curves from the four sites of Fig. 2(a). The very hard magnetization of site J is explained by the presence of hematite as the main magnetic mineral, since the Curie Point is above 650 °C. The unusually steep thermal demagnetization curve of site O is a characteristic possessed by many other sites.

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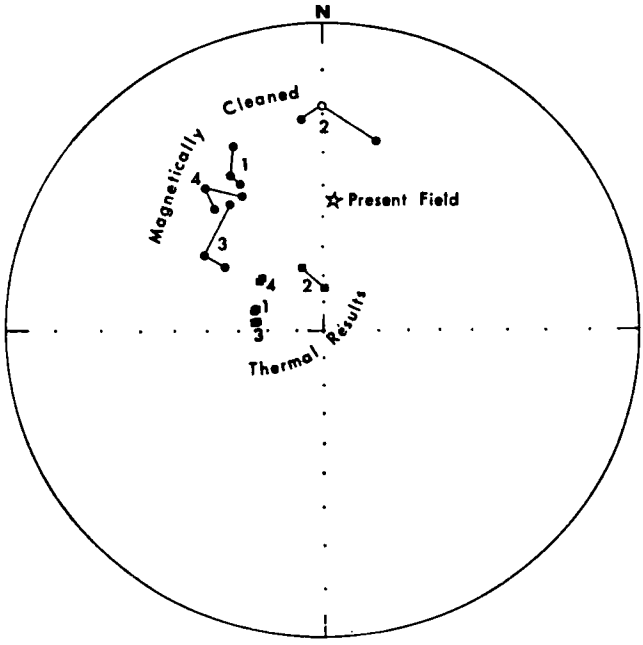


FIG. 3. Directions of magnetization of specimens from site U after magnetic (circles) and thermal (squares) cleaning. Specimens from the same sample are linked, and the present geomagnetic field is represented by the star. Stereographic projection—solid symbols lower hemisphere, open symbols upper hemisphere.

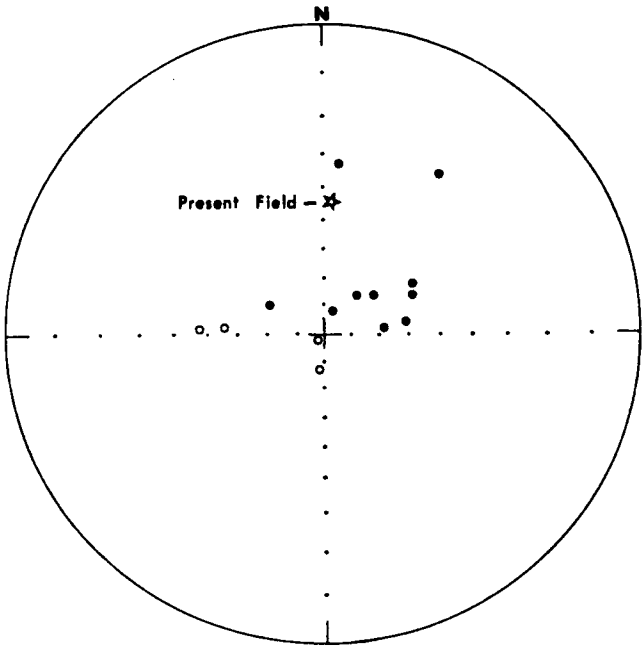


FIG. 4. Mean directions of magnetization after cleaning from the 14 accepted sites. Stereographic projection, symbols as in Fig. 3.

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Except for sites G, J, O and U, sample directions are more closely grouped after magnetic cleaning than after thermal cleaning so that the former method has been preferred. At sites G and J, neither the mean directions nor the precisions are significantly different using the two methods; the magnetically cleaned directions have been preferred. Specimens from site O exhibited a marked increase in susceptibility during thermal demagnetization, and individual magnetometer readings were subject to small fluctuations during the course of measurement. The magnetically cleaned result is therefore preferred. This increase in susceptibility during thermal treatment is a feature of many other sites, some of which gave wildly fluctuating readings after heating to only 300 °C. Finally, the thermal results from site U (Fig. 3) give a mean direction further from the direction of the present geomagnetic field and closer to that of the other cleaned sites. Therefore in this case the thermally cleaned result is preferred.

The 14 accepted site mean directions after cleaning group about two positions (Fig. 4), one directed fairly steeply downwards in the first quadrant, the other upwards

Table 3

Summary of Phanerozoic palaeomagnetic data from Australia

No.	Formation	Age (K–Ar date)	Pole position		A_{95}	Reference
Au 1	Antrim Plateau Volcanics	€l	9 S	20 W	17	This paper
Au 2	Jinduckin Formation	Ol	16 S	29 E	17	Luck (1970)
Au 4	Mugga Mugga Porphyry	Su (423 ± 9* My)	80 S	20 E	7	Briden (1966)
Au 5a	Lower Kuttung Lavas	Cl–m	84 S	46 E	12	Irving (1966)
Au 5b	Paterson Toscanite	Cm–u	73 S	147 E	5	Irving (1966)
Au 5c	Main Glacial Stage	Cu	53 S	148 E	11	Irving (1966)
Au 5d	Rocky Creek Conglomerate	Cu	52 S	138 E	17	Irving (1966)
Au 5e	Currabubulla Formation	Cu	43 S	135 E	24	Irving (1966)
Au 6a	Lower Marine Basalt	Pl	46 S	122 E	9	Irving & Parry (1963)
Au 6b	Upper Marine Latites	Pm–u (248 My)	44 S	132 E	12	Irving & Parry (1963)
Au 6c	Moonbi Lamprophyre	P	35 S	128 E	20	Irving (1966)
Au 6d	Milton Monzonite	Pu (240 My)	32 S	170 E	24	Robertson (1964)
Au 7a	Narrabeen Choc. Shales	Trl	49 S	160 E	14	Irving (1963)
Au 7b	Brisbane Tuff	Trm	57 S	143 E	11	Robertson (1963)
Au 8a	Gibraltar Microsyenite	Jl (178 My)	41 S	146 E	24	Boesen *et al.* (1961)
Au 8b	Prospect dolerite	Jm (168 My)	51 S	151 E	13	Boesen *et al.* (1961)
Au 8c	Tasmanian dolerites	Jm (167 My)	51 S	160 E	6	Irving (1963)
Au 8d	Noosa Heads Complex	Ju (140 My)	36 S	132 E	24	Robertson (1963)
Au 8e	Gingenbullen dolerite	J	53 S	144 E	15	Boesen *et al* (1961)
Au 9a	Cygnet Complex	K (98 My)	50 S	158 E	10	Robertson & Hastie (1962)
Au 9b	Mt. Dromedary Complex	K (93 My)	56 S	138 E	9	Robertson (1963)
Au 10a	Older Volcanics Victoria	Palaeocene–Eocene	63 S	140 E	16	Mumme (1963)
Au 10b	Barrington Volcano	Eocene (52 My)	70 S	126 E	5	Wellman *et al.* (1969)
Au 10c	Liverpool Volcano	Oligocene (34 My)	71 S	96 E	5	Wellman *et al.* (1969)
Au 11a	Main Range and Tweed Volcanoes	Miocene (21–24 My)	75 S	112 E	4	Wellman (1970)
Au 11b	S.E. Queensland Igneous Rocks	Miocene (21–24 My)	78 S	86 E	12	Robertson (1966)
Au 11c	Nandewar Volcano	Miocene (17·5 My)	78 S	84 E	5	Wellman *et al.* (1969)
Au 11d	Newer Volcanics Victoria	Plio-Pleist. (4·5–0·5 My)	83 S	123 W	8	McDougall *et al.* (1966)

*Rb–Sr isochron age (Bofinger *et al.* 1970) using decay constant $1{\cdot}39 \times 10^{-11}$ yr^{-1}.

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in the third quadrant. These latter four sites are reversely magnetized. In Table 2, site pole positions are noted against each of the accepted cleaned results and from these have been calculated a mean pole position for the period of extrusion of the Volcanics.

Phanerozoic polar-wander path for Australia

The new results determined in this and the previous paper (Luck 1970), enable us to complete the Phanerozoic polar-wander path for Australia by extending it back through the Ordovician to Lower Cambrian times. In Table 3, therefore, we have listed all Australian results for the Phanerozoic that have been derived using cleaning techniques and for which there is good age control. In some cases ages have been determined from either K–Ar or Rb–Sr decay, as indicated.

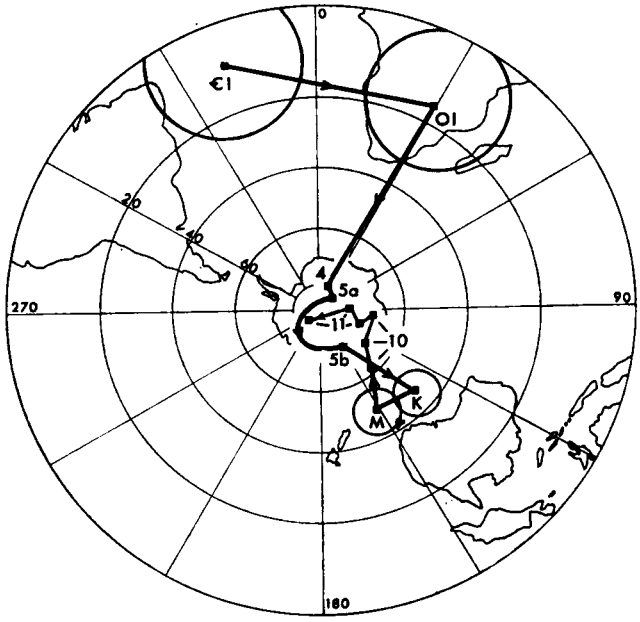


FIG. 5. Phanerozoic polar-wander path for Australia. The poles are numbered as in Table 3 and circles of 95 per cent confidence have been indicated for the Antrim Plateau Volcanics (€1), the Jinduckin Formation (O1), the Mean Kiaman (K) and Mesozoic (M) poles (Table 4).

The complete Phanerozoic polar-wander path is illustrated in Fig. 5, which shows a 40° difference between the Lower Cambrian and Lower Ordovician poles. Because of fairly large errors (17° in each case) associated with these poles, the real difference may be smaller than that indicated, a detail which should be clarified by work currently in progress. However, the difference is significant ($40° \pm 24°$) and it does appear that the 'quasi-static interval', designated by Briden (1967) for the lower Palaeozoic, contains real polar movement, which could have been quite large.

The most important feature revealed by the new results is a polar transition of 60° which takes place between the Lower Ordovician and the Upper Silurian. This may be compared with the similar transitions which occur in data from Africa

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(McElhinny *et al.* 1968) and South America (Creer 1965). A further rapid transition occurs during the Carboniferous, as documented by Irving (1966), and a similar transition has also been proposed for Africa (McElhinny & Opdyke 1968). Following this, the pole remains in a virtually unchanged position just south of the Australian Bight from the Upper Carboniferous to the mid-Cretaceous (100 My ago). Irving (1966) has proposed that no differences can be claimed between any of these poles and therefore averaged the 15 results he listed to form the Mean Kiaman and Mesozoic Pole for Australia. One further result may now be added to Irving's list, that for the Milton Monzonite, whose age has recently been determined as 240 My (Richards, personal communication). This places it in the uppermost Permian, very close to the Permo-Triassic boundary of 235 My (Webb & McDougall 1967). A closer look at the 16 results listed in Table 3 between the Upper Carboniferous and mid-Cretaceous shows that the six poles older than the Milton Monzonite tend to have longitudes to the west of the other 10 members of this group. The feature about these early six poles is that they lie within the Kiaman Magnetic Interval (Irving & Parry 1963), whereas the Milton Monzonite has an age so close to the Permo-Triassic boundary that it might well be assumed to be post-Kiaman. Such grouping may have some significance, and this will now be tested.

Table 4

Means for groups of Australian Poles (*see Table* 3)

Group	Pole Nos.	N	R	K	Lat. S	Long. E	A_{95}
Mean Kiaman	5c, d, e; 6a, b, c	6	5·9395	82·6	45·8	133·3	7·4
Mean Mesozoic	6d; 7a, b; 8a, b, c, d, e; 9a, b	10	9·8105	47·5	48·2	150·6	7·1
Mean Kiaman & Mesozoic	5c, d, e; 6a, b, c, d; 7a, b; 8a, b, c, d, e; 9a, b	16	15·6685	45·2	47·6	143·9	5·5

N, No. of poles; R, Resultant Vector; K, precision; A_{95}, radius of circle of 95 per cent confidence about mean position given.

The parameters of the mean Kiaman and mean Mesozoic poles are given in Table 4. They have also been plotted with their 95 per cent confidence limits in Fig. 5. The circles of confidence overlap, whilst the angular distance between the two group means is 12·5°. This does not necessarily mean that they are not significantly different. To test this, the statistic

$$(N-2)\frac{(R_1+R_2-R)}{(N-R_1-R_2)}$$

may be compared with F-ratio tables with 2 and $2(N-2)$ degrees of freedom (Watson & Irving 1957). In this case the $N_1 = 6$ group of Kiaman poles has resultant R_1, whilst the $N_2 = 10$ group of Mesozoic poles has resultant R_2. When all $N = 16$ results are combined, the resultant has length R. From this we obtain

$$F_{2,28} = 4{\cdot}56^*$$

which is significant, since the 5 per cent significance point is 3·34. Therefore there is justification for considering that the Kiaman and Mesozoic poles form two distinct groups.

The polar-wander path is then completed by the transition of the pole from its Mesozoic position to the present position of the South Pole as outlined by Wellman, McElhinny & McDougall (1969). A further result from the Tertiary has also been

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included. Because this path terminates at the present South Pole, it can be considered as the path of the South Pole during the Phanerozoic. During the 600 My period the pole has moved through nearly 180°, most of which has taken place during short rapid transitions at a rate very much faster than the average of 0·3° per million years or about 3 cm yr^{-1}.

From the palaeomagnetic poles listed in Table 3, the palaeolatitude of any place in Australia can be determined. Consequently the variation in latitude of Canberra during the whole of the Phanerozoic has been plotted in Fig. 6. Since the angular distance between Australia and its Lower Cambrian pole in Fig. 5 (inferred to be the geographic South Pole) is greater than 90°, this implies that Australia was situated in the northern hemisphere during Cambrian times.

Australia and Africa during the Palaeozoic

It has been shown (McElhinny 1967) that the polar-wander paths for Africa and South America coincide for virtually the whole of the Palaeozoic when the South Atlantic is closed. The position of Australia could only be inferred from upper Palaeozoic results (McElhinny & Opdyke 1968) by matching the Carboniferous transitions observed in both Africa and Australia. We can now see whether the polar paths for these two continents match for the whole of the Palaeozoic. If so, then this will tend to confirm the interpretation of Permian and Carboniferous results from Africa made by McElhinny & Opdyke (1968).

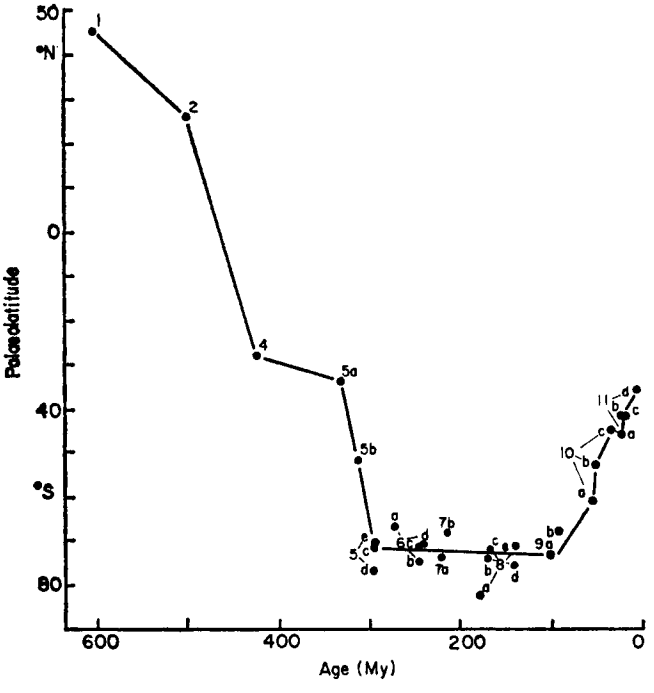


FIG. 6. Variation in palaeolatitude of Canberra during the Phanerozoic. The points are numbered as in Table 3.

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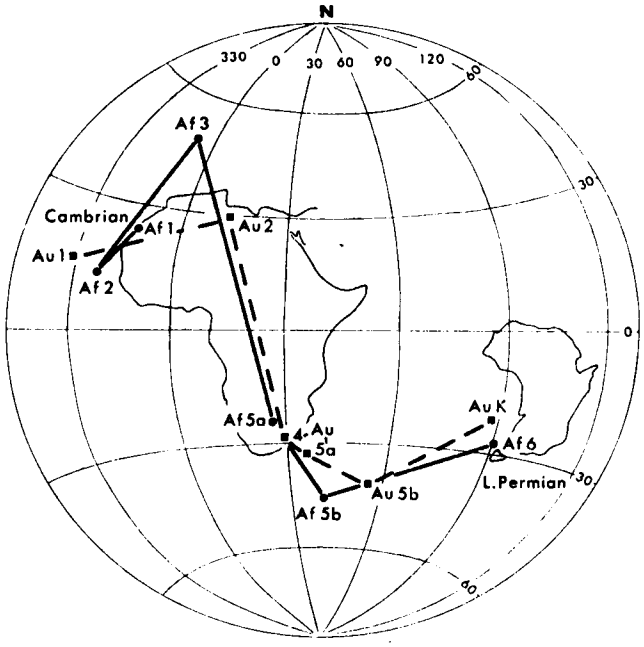


FIG. 7. Comparison of African and Australian Palaeozoic polar-wander paths. The position of Australia has been adjusted so that the best fit of the two paths is obtained. Equal area meridional projection with Africa in its present co-ordinates. Australian poles (squares) are numbered as in Table 3 with the Mean Kiaman pole (K) from Table 4. African poles (circles) are numbered as in Table 5.

Table 5

Summary of Palaeozoic results from Africa (after McElhinny et al. 1968)

Pole No.	Formation	Age (K–Ar or Rb–Sr date)	Pole position		A_{95}
Af 1	Ntonya Ring Structure	L. Cambrian (600 My)	28 N	15 W	2
Af 2	Hook Intrusives	L. Ordovician (500 My)	15 N	26 W	36
Af 3	Table Mountain Series Shales	Ordovician*	50 N	11 W	4
Af 5a	Dwyka Varves	L. Carboniferous	26 S	26 E	10
Af 5b	K3 Galula redbeds	U. Carboniferous	46 S	40 E	8
Af 6	K3 Ketewaka–Songwe redbeds	L. Permian	27 S	89 E	16

* Evidence for the change in age of this formation is given by Cocks *et al.* (1969).

Palaeozoic results for Africa are summarized in Table 5 and these are combined with those listed for Australia in Table 3 to construct the relative positions of Africa and Australia as shown in Fig. 7. The best match of the paths for the whole of the Palaeozoic is obtained when Australia is placed a little further north with respect to Africa than previously proposed by McElhinny (1967). This of course implies that the relative positions of these two continents remained unchanged for the whole of the Palaeozoic.

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